

Descriptive Statistics/Questions with Linear Regression

August 31 and September 5 2017

Outline

Describing Univariate and Multivariate Populations

Description with Linear Regression

Description with Multiple Regression

Plan for Today and Tuesday

- Descriptive Inference is about using samples to answer questions about populations.
- There are many possible questions we might have about populations.
- To simplify the discussion, we will start by analyzing the case where we observe the entire population (i.e., a census) and no sampling is necessary.
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Example: Epstein and Mershon SCOTUS data

- Data on 27 justices from the Warren, Burger, and Rehnquist courts (can be interpreted as a census)
- Percentage of votes in liberal direction for each justice in a number of issue areas
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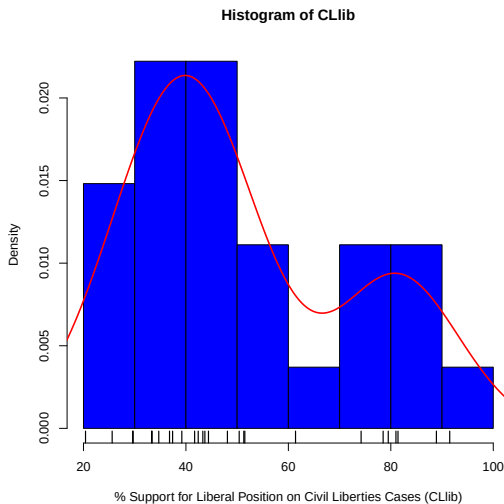
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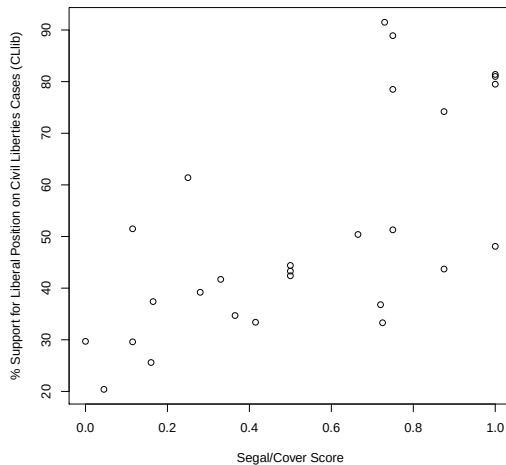
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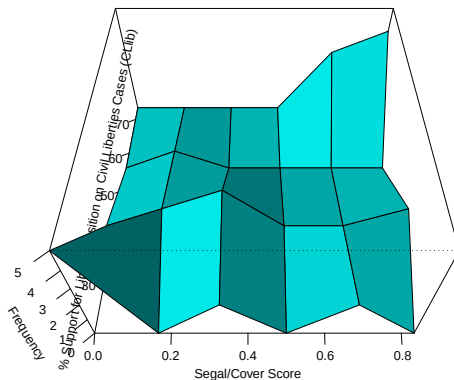
Describing Univariate Data



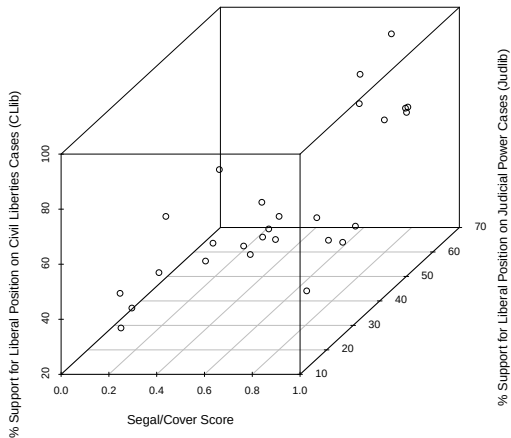
Describing Bivariate Data



Describing Bivariate Data



Describing Trivariate Data



Summarization

- Even for univariate data, we may want to focus on one or two characteristics of the population instead of describing it in its entirety.
- We typically focus on measures of center and spread.
- We will discuss the rational for this over the coming weeks.

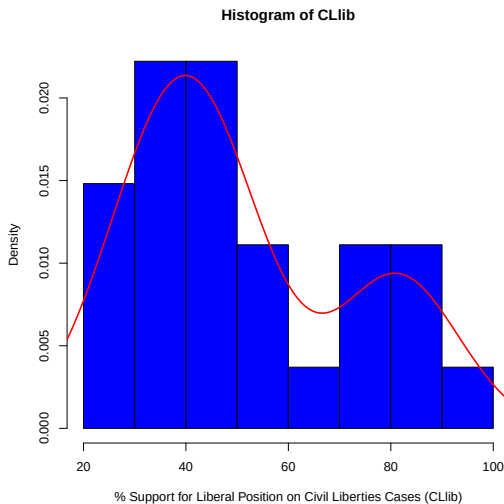
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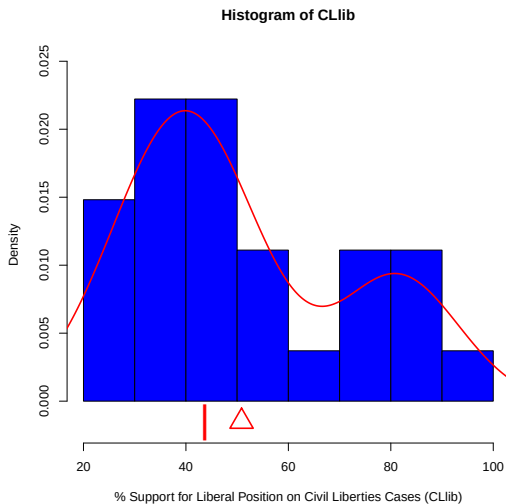
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Univariate Activity



Univariate Activity Solution



Why Do We Focus on Means?

- Population means ($\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$) often provide a “good” summary of the center of the data (and it is relatively easy to tell when they provide bad summaries).
- The accuracy of means is relatively easy to describe.
- Randomized experiments only identify average causal effects (as do regression and matching when the appropriate assumptions hold).

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The Mean as a Least Squares Summary

Suppose we want to pick a single number (m) for the middle of the data that summarizes all the values for y , by minimizing the sum of squared residuals (i.e., least squares).

$$SSR(\tilde{m}) = \sum_{i=1}^N (y_i - \tilde{m})^2.$$

One way to calculate the least squares estimator

1. Calculate the derivative of SSR with respect to $\tilde{m}$
2. Set the derivative equal to 0
3. Solve for m

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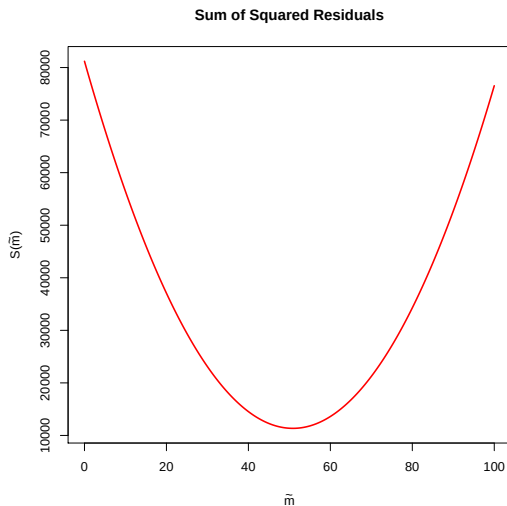
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The Objective Function for SSR CLlib

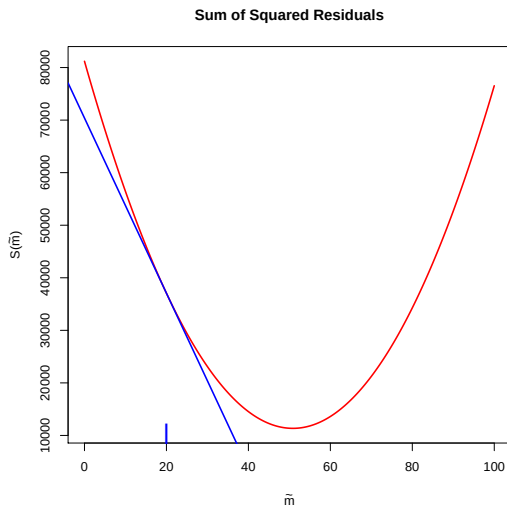


1.

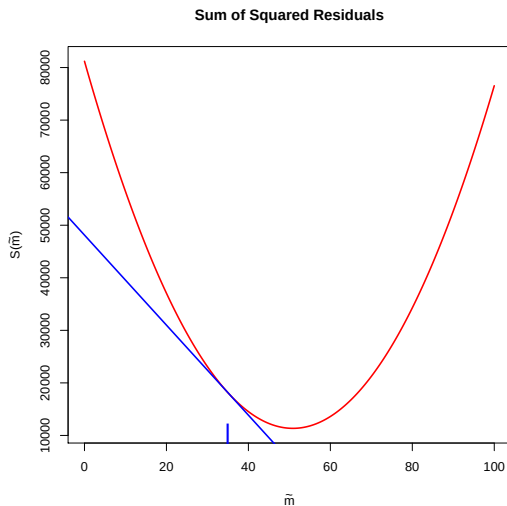
$$\begin{aligned} SSR(\tilde{m}) &= \sum_{i=1}^N (y_i - \tilde{m})^2 \\ &= \sum_{i=1}^N (y_i^2 - 2y_i\tilde{m} + \tilde{m}^2) \end{aligned}$$

$$\frac{\partial SSR(\tilde{m})}{\partial \tilde{m}} = \sum_{i=1}^N (-2y_i + 2\tilde{m})$$

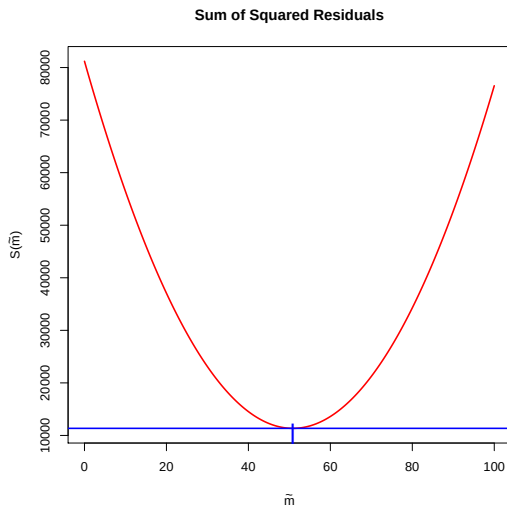
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2.

$$0 = \sum_{i=1}^N (-2y_i + 2m)$$

3.

$$m = \frac{1}{N} \sum_{i=1}^N y_i$$

Therefore, the population mean is a least squares summary.

Variance and Standard Deviation

- Mean is the least squares predictor in terms of SSR.
- How small is “least” (i.e., how much spread around the mean)?
- Variance and Standard Deviation are useful transformations of SSR for the mean.

Population Variance

$$\begin{aligned} S^2 &= \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N - 1} \\ &= \frac{SSR}{N - 1} \end{aligned}$$

Population Standard Deviation

$$S = \sqrt{S^2}$$

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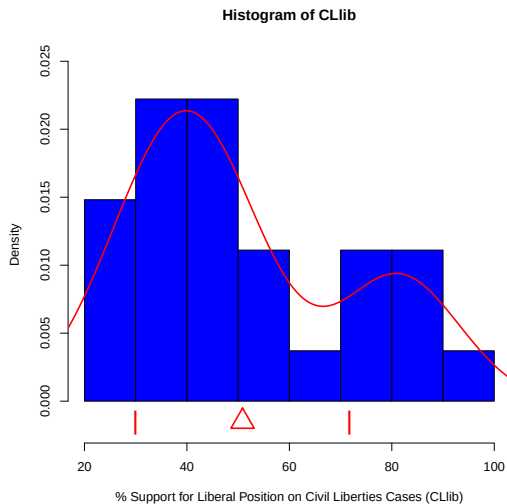
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Mean and Standard Deviation



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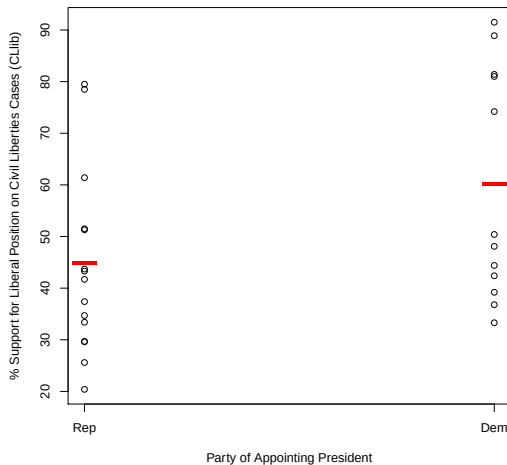
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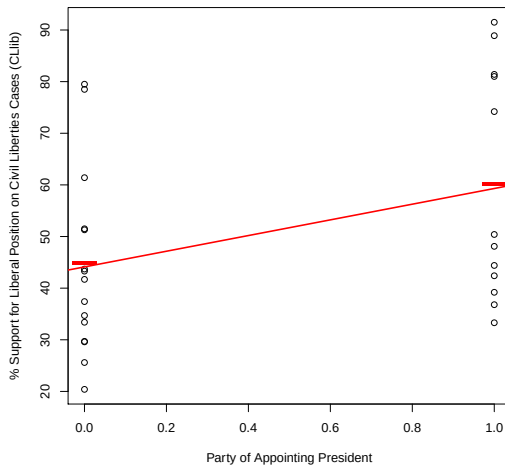
Conditional Means

- In this class, we focus on a single “outcome” variable.
- All additional variables adjust how we consider the outcome, often by conditioning.
- Typically, we consider the mean of y conditional on values of other variables.

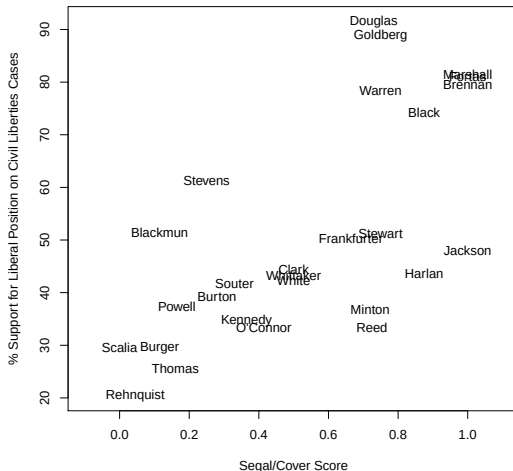
Mean CLlib Conditional on Party



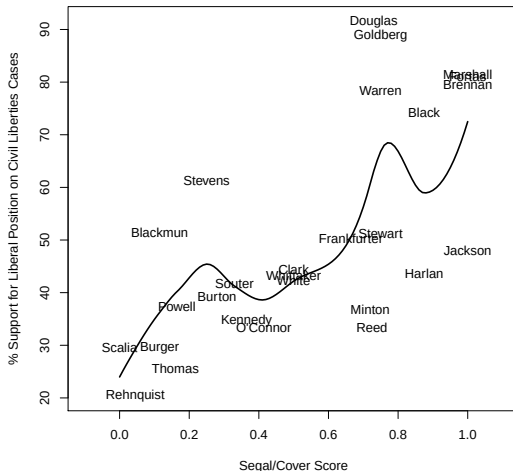
Linear Conditional Mean Function (LCMF)



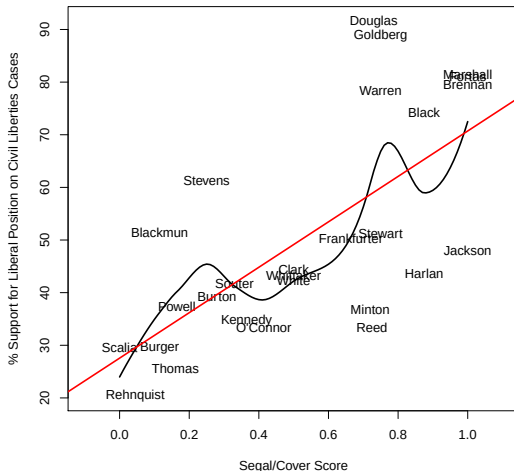
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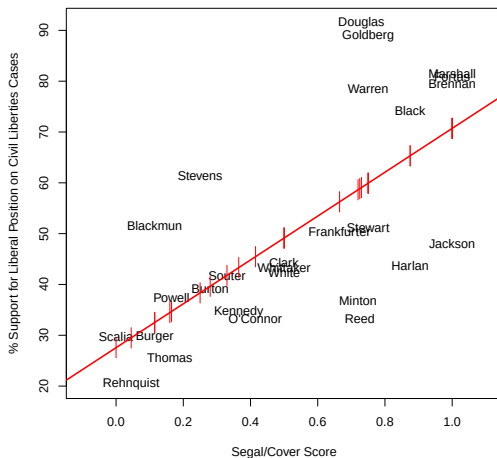


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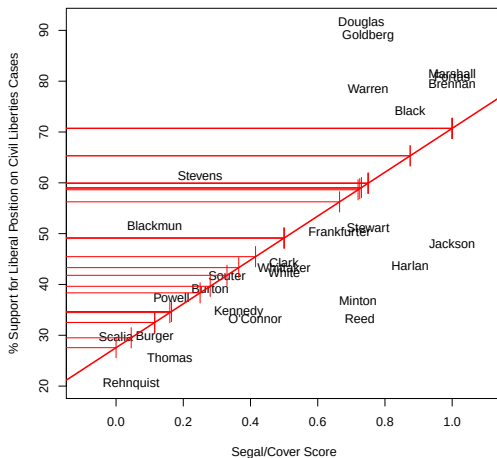
Fitted Values and Residuals

- Fitted Values from LCMF: $\hat{y}_i = B_0 + B_1 x_i$ for $i = 1, \dots, N$
- Residuals from Fitted Values: $e_i = y_i - \hat{y}_i$ for $i = 1, \dots, N$



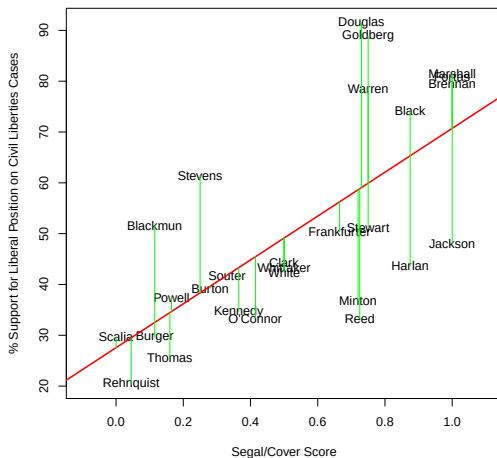
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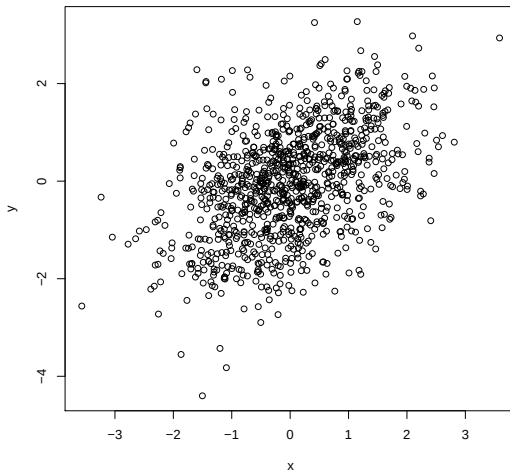


Minimizing the Residuals

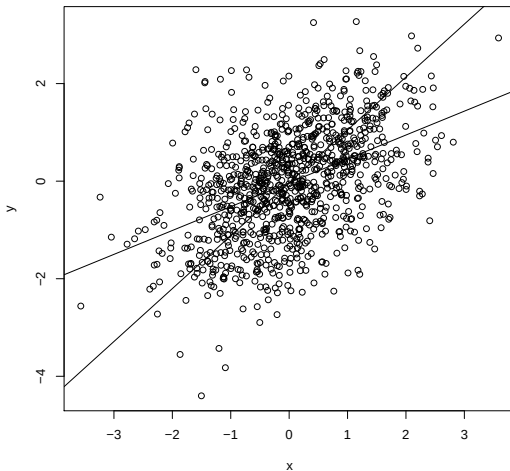
- Let $\tilde{B}_0$ and $\tilde{B}_1$ be possible values for B_0 and B_1 respectively,
- *Least Squares (LS) regression:*

$$(B_0, B_1) = \underset{\tilde{B}_0, \tilde{B}_1}{\operatorname{argmin}} \sum_{i=1}^N (y_i - \tilde{B}_0 - x_i \tilde{B}_1)^2$$

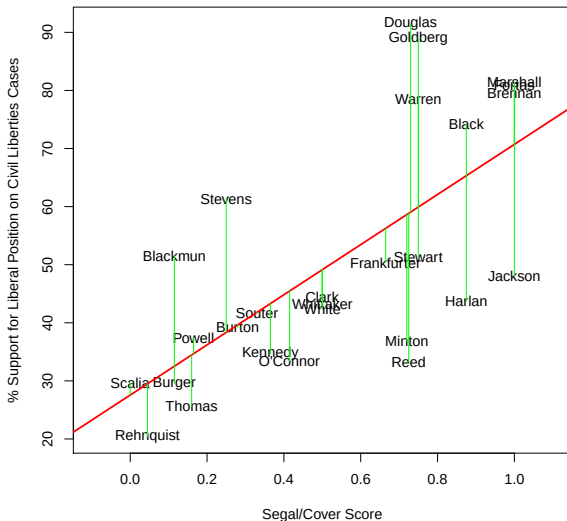
Simple Regression Activity



Simple Regression Activity



Choosing Slope and Intercept to Minimize SSR



Deriving the Linear Least Squares Estimator

Let $\tilde{B}_0$ and $\tilde{B}_1$ be possible values for B_0 and B_1 respectively, and

$$S(\tilde{B}_0, \tilde{B}_1) = \sum_{i=1}^N (y_i - \tilde{B}_0 - x_i \tilde{B}_1)^2.$$

1. Take partial derivatives of S with respect to $\tilde{B}_0$ and $\tilde{B}_1$.
2. Set each of the partial derivatives to 0
3. Substitute B_0 and B_1 for $\tilde{B}_0$ and $\tilde{B}_1$ and solve for B_0 and B_1

Useful Results

The partial derivatives imply that

$$\sum_{i=1}^N x_i e_i = 0$$

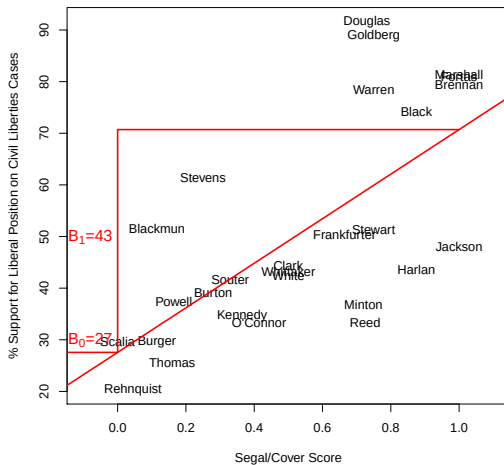
$$\sum_{i=1}^N \hat{y}_i e_i = 0$$

and the **LS LCMF**:

$$B_0 = \bar{y} - B_1 \bar{x}$$

$$B_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

Descriptive Interpretation



Variance and Regression

- Again, we want to assess the “leastness” of least squares (i.e., the spread of the data around the line).
- The standard error of the regression, $\sqrt{\frac{\sum_{i=1}^N e_i^2}{N-2}}$, is analogous to the standard deviation.
- There are other approaches to this question: correlation and coefficient of determination.

Covariance

The population variances for X and Y can be written as:

$$\begin{aligned} S_x^2 &= \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1} \\ &= \frac{\sum_{i=1}^N (x_i - \bar{x})(x_i - \bar{x})}{N - 1} \end{aligned}$$

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This quantity is the **covariance**.

Correlation

The covariance does not have units that are easy to interpret, so we can scale it by the standard deviations of X and Y to obtain the **correlation**:

$$R = \frac{S_{xy}}{S_x S_y} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}}$$

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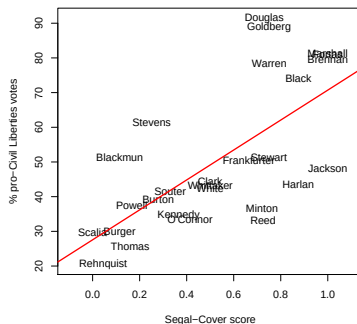
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Analysis of Variance for the Regression

How well do the Segal-Cover scores predict support for civil liberties in relation to the average support for civil liberties?

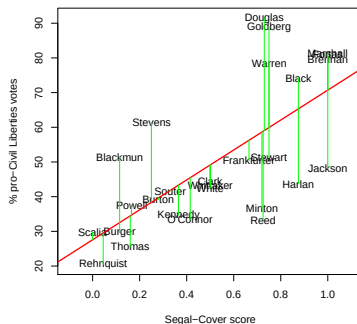
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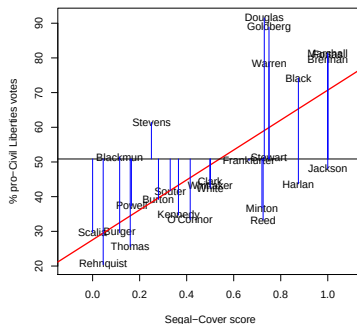
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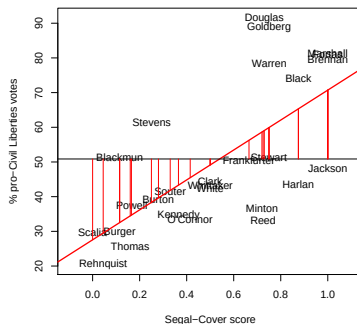
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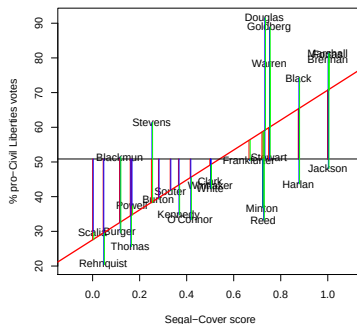
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$$\sum_{i=1}^N (y_i - \bar{y})^2 = \sum_{i=1}^N \{e_i + (\hat{y}_i - \bar{y})\}^2$$

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Coefficient of Determination

We can divide each side by the TSS:

$$\frac{SSR}{TSS} + \frac{RegSS}{TSS} = \frac{TSS}{TSS}$$

$$\frac{SSR}{TSS} + \frac{RegSS}{TSS} = 1$$

$$\frac{RegSS}{TSS} = 1 - \frac{SSR}{TSS} = R^2$$

R^2 is a measure of how much of the variation in Y is accounted for by X .

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Usefull Identities

$$R^2 = (R)^2$$

$$B_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

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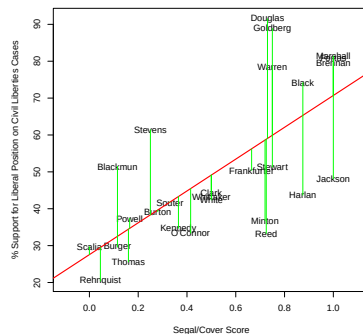
$$\begin{aligned} B_1 &= \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} \\ &= \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}} \cdot \frac{\sqrt{\frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N-1}}}{\sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1}}} \end{aligned}$$

Usefull Identities

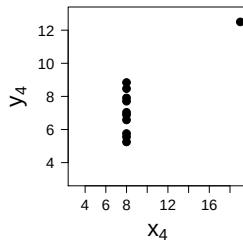
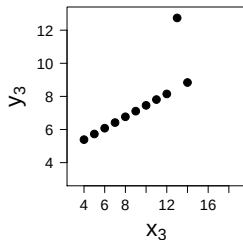
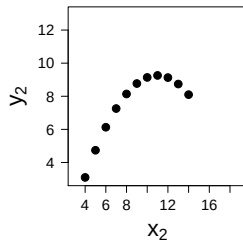
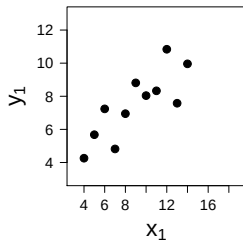
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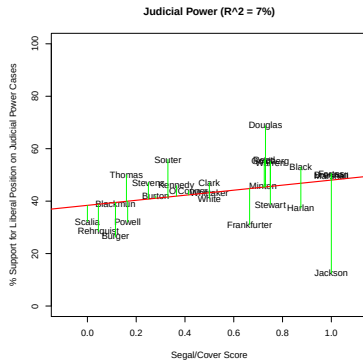
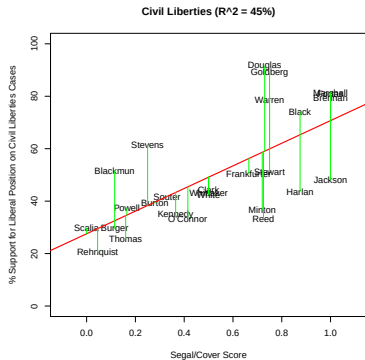
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- $R^2 = 0.45$
- $R = \sqrt{R^2} = 0.67$
- $S_x = 0.33$
- $S_y = 20.9$
- $B_1 = R \frac{S_y}{S_x} = 43.15$



Anscombe's Quartet





Outline

Describing Univariate and Multivariate Populations

Description with Linear Regression

Description with Multiple Regression

More Than One Conditioning Variable

- Often we want averages conditional on more than one variable.
- As with continuous conditioning variables, there are many ways to do this.
- In this class we will look at two.

Multiple Regression with 2 Conditioning Variables

- Fitted Values from the LCMF $\hat{y}_i = B_0 + B_1 x_i + B_2 z_i$ for $i = 1, \dots, N$
- Residuals from Fitted Values: $e_i = y_i - \hat{y}_i$ for $i = 1, \dots, N$

Again, we choose B_0 , B_1 , and B_2 , to minimize the SSR, where

$$SSR = \sum_{i=1}^N e_i^2$$

but the definition of e_i changes with the inclusion of more variables.

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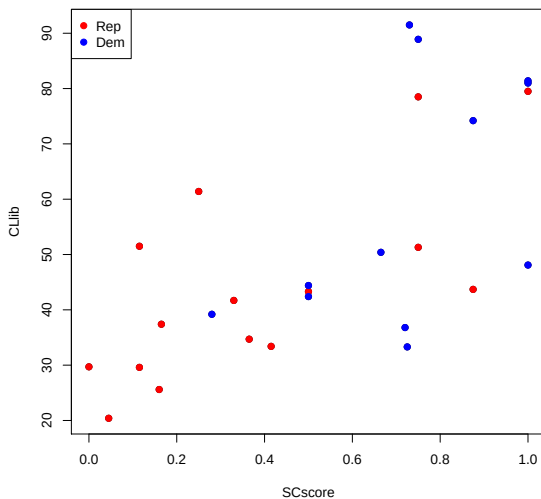
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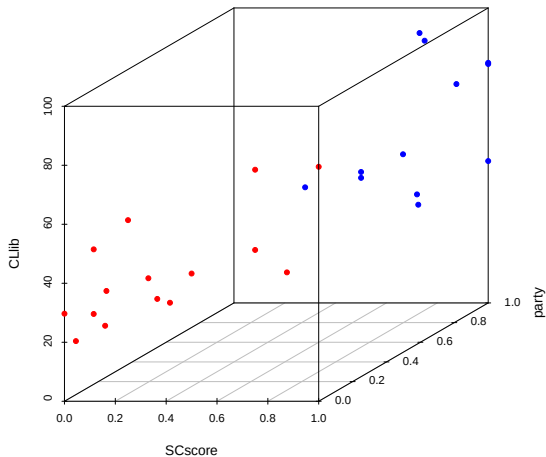
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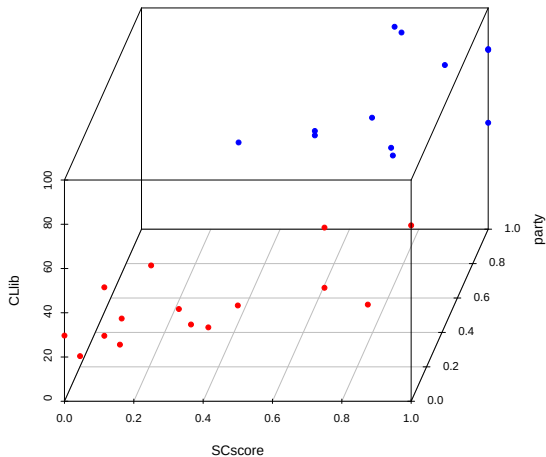
One Binary and One Continuous Conditioning Variable



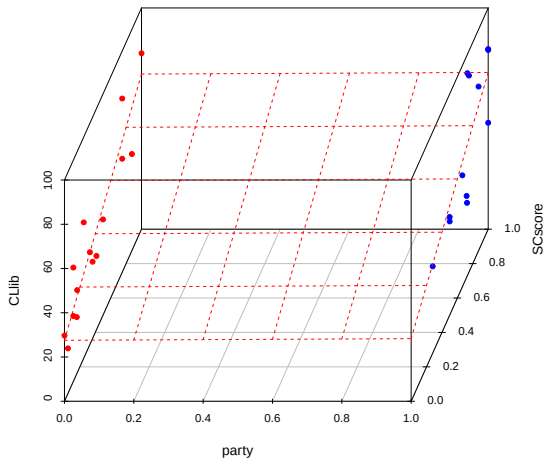
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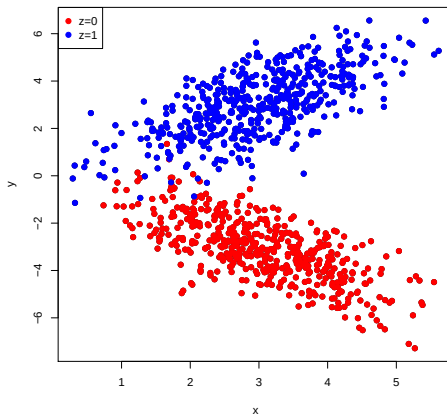
Regression Plane



Multiple Regression Activity

Draw the two regression lines (i.e., edges of the regression plane) implied by

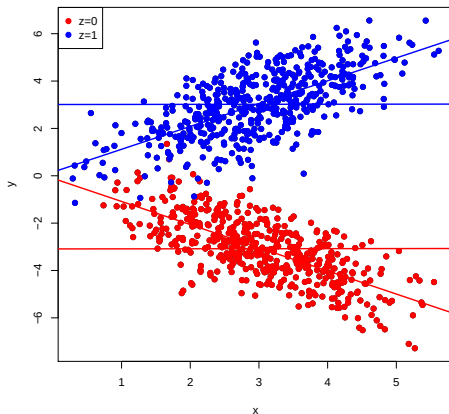
$$\hat{y} = B_0 + B_1x + B_2z$$



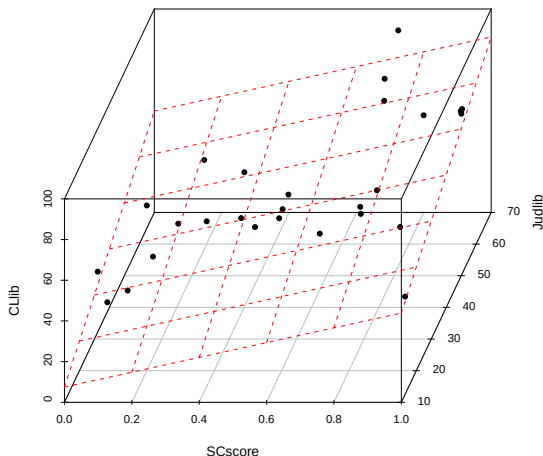
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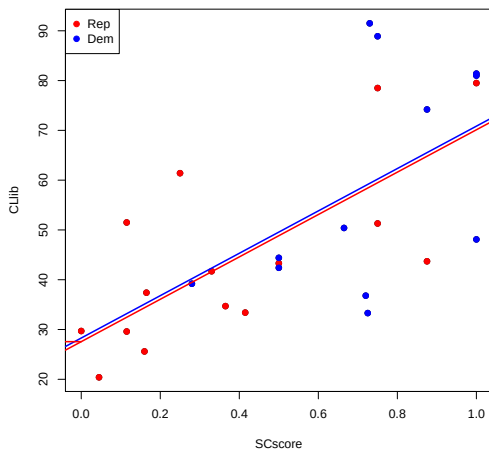


Regression Plane for Two Continuous Conditioning Variables



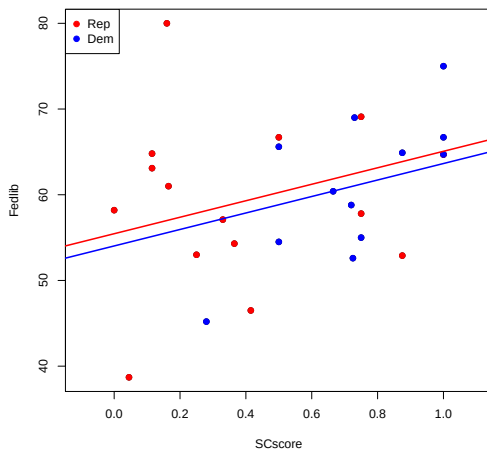
Descriptive Interpretation of Additive Regression

$$\widehat{CLib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i$$



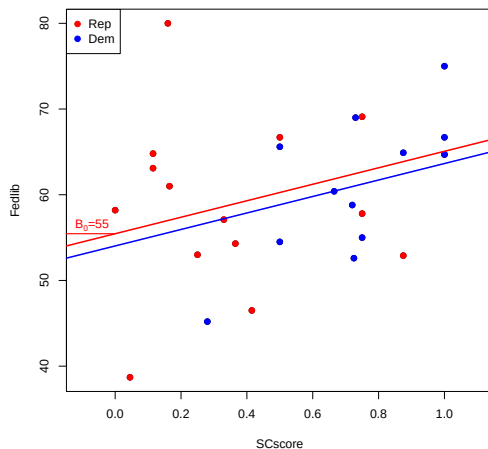
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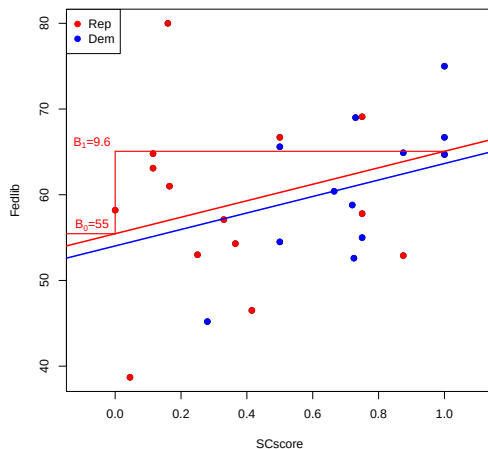
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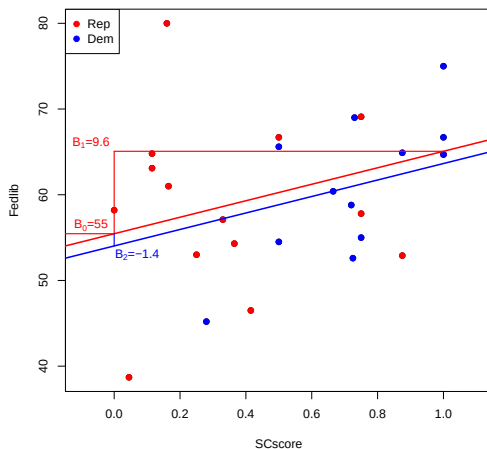
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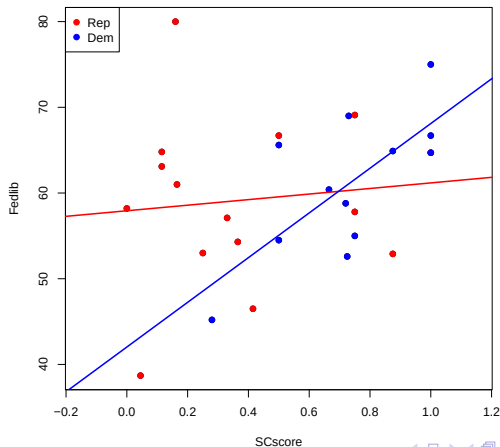
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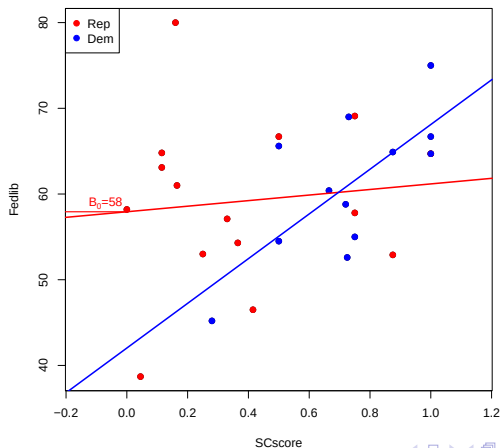
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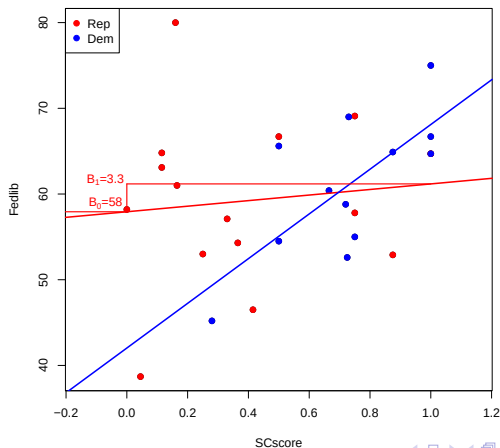
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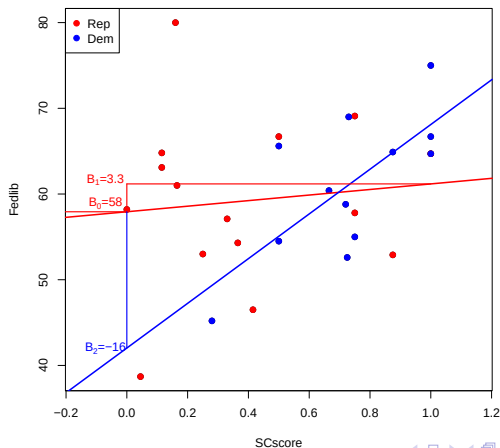
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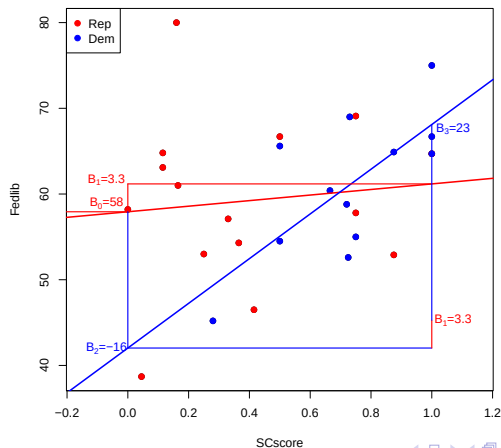
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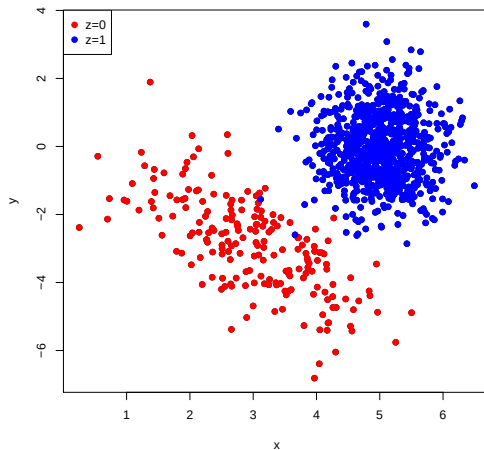
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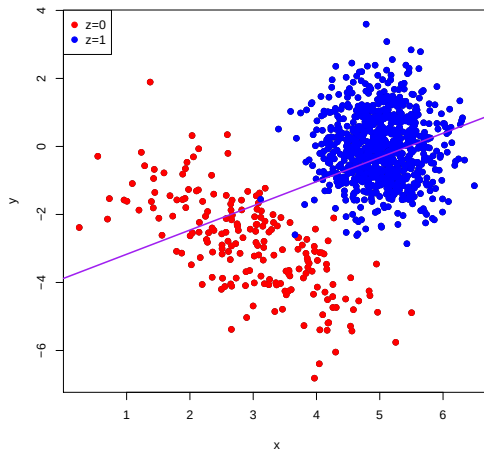
Variance and Multiple Regression

- As before, we want to assess the “leastness” of least squares (i.e., the spread of the data around the plane or twisted plane).
- The standard error of the regression, $\sqrt{\frac{\sum_{i=1}^N e_i^2}{N-k}}$, now accounts for the number of B ’s in the model (where k is the number of B ’s).
- R^2 uses the same formula $1 - \frac{SSR}{TSS}$, however the inclusion of additional variables will automatically reduce the SSR.

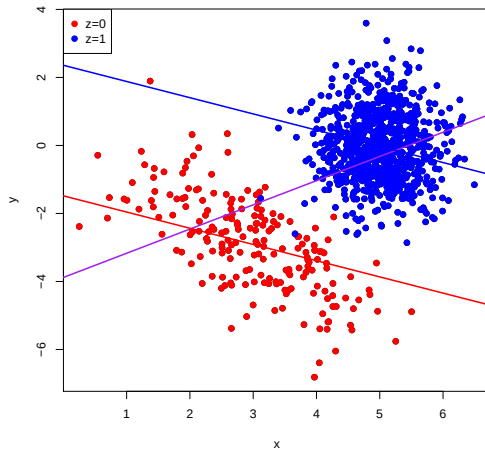
Summary Activity



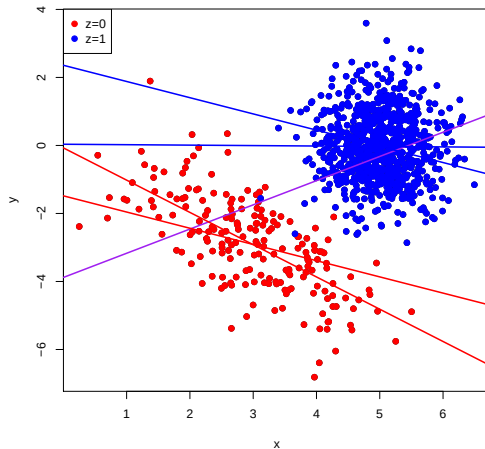
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